Kubernetes and Istio

Comprehensive Technical Documentation

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12 - Istio security

Istio provides the following security features:

- **Mutual pods authentication** via Mutual TLS: see reference
- **End-user authentication**: configurable JSON Web Token verification per path see reference
- **Filter HTTP traffic**: allow only traffic with some characteristics between pods see reference
- **Authorization**: give different kubernetes accounts different permissions see reference

Authentication Policy
1. Verifies mTLS & JWT
Target Kubernetes Pod
✅ Allowed
📦 Application Container
⚠️ External Attacker / Compromised Pod
Malicious HTTP Request (e.g., missing JWT, wrong path)
🛡️ Envoy Proxy (Sidecar)
Authorization Policy
2. Verifies RBAC & HTTP Rules
❌ Denied (401 / 403)
Traffic Dropped